Complex Number: Questions and Full Solutions

Question 01

English

For each real number θ, let $w(\theta) = (3+4i)e^{i\theta} + (7-i)e^{-i\theta}$, where $i = \sqrt{-1}$. As θ varies over $\mathbb{R}$, the point representing $w(\theta)$ in the Argand plane traces an ellipse. Then the slope of the major axis of this ellipse is

(A) $-(\sqrt{2}+1)$
(B) 1
(C) $\sqrt{2}-1$
(D) $\dfrac{4}{3}$

Only one option is correct.

हिन्दी

प्रत्येक वास्तविक संख्या θ के लिए माना कि $w(\theta) = (3+4i)e^{i\theta} + (7-i)e^{-i\theta}$ है, जहाँ $i = \sqrt{-1}$ है। जब θ सम्पूर्ण $\mathbb{R}$ पर परिवर्तित होता है, तब आर्गण्ड तल में $w(\theta)$ को निरूपित करने वाला बिन्दु एक दीर्घवृत्त अनुरेखित करता है। तब इस दीर्घवृत्त के दीर्घ अक्ष की प्रवणता है

(A) $-(\sqrt{2}+1)$
(B) 1
(C) $\sqrt{2}-1$
(D) $\dfrac{4}{3}$

केवल एक विकल्प सही है।

Solution

Let $\alpha = \arg(3+4i)$ and $\beta = \arg(7-i)$. Then $3+4i = 5e^{i\alpha}$ with $\tan\alpha = \dfrac{4}{3}$, $\alpha \in \left(0, \dfrac{\pi}{2}\right)$, and $7-i = 5\sqrt{2}\,e^{i\beta}$ with $\tan\beta = -\dfrac{1}{7}$, $\beta \in \left(-\dfrac{\pi}{2}, 0\right)$.

By the compound-angle formula, $\tan(\alpha+\beta) = \dfrac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta} = \dfrac{\frac{4}{3} - \frac{1}{7}}{1 + \frac{4}{21}} = \dfrac{25/21}{25/21} = 1$.

Since $\alpha+\beta \in \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$, $\alpha+\beta = \dfrac{\pi}{4}$. (The same follows from $(3+4i)(7-i) = 25+25i$.)

Now $w(\theta) = 5e^{i(\theta+\alpha)} + 5\sqrt{2}e^{-i(\theta-\beta)}$. Take out the factor $e^{i(\alpha+\beta)/2}$ and put $\psi = \theta + \dfrac{\alpha-\beta}{2}$:

$w = e^{i(\alpha+\beta)/2}\left(5e^{i\psi} + 5\sqrt{2}e^{-i\psi}\right)$.

$5e^{i\psi} + 5\sqrt{2}e^{-i\psi} = (5+5\sqrt{2})\cos\psi + i(5-5\sqrt{2})\sin\psi$.

As θ runs over $\mathbb{R}$, so does ψ, and the point $X+iY$ with $X = (5\sqrt{2}+5)\cos\psi$, $Y = -(5\sqrt{2}-5)\sin\psi$ traces the ellipse $\dfrac{X^2}{(5\sqrt{2}+5)^2} + \dfrac{Y^2}{(5\sqrt{2}-5)^2} = 1$ with its major axis along the X-axis.

Multiplying by $e^{i(\alpha+\beta)/2} = e^{i\pi/8}$ rotates this ellipse through $\dfrac{\pi}{8}$ about the origin, so the major axis of the locus of w makes the angle $\dfrac{\pi}{8}$ with the real axis.

$$\text{Slope} = \tan\frac{\pi}{8} = \frac{\sin(\pi/4)}{1+\cos(\pi/4)} = \frac{1/\sqrt{2}}{1+1/\sqrt{2}} = \frac{1}{\sqrt{2}+1} = \sqrt{2}-1.$$

(The minor axis has slope $-(\sqrt{2}+1)$; the direction $\dfrac{\pi}{4}$ of $\alpha+\beta$ itself would give the wrong slope 1.)

Correct Option: (C)

Question 02

English

Let S be the set of all complex numbers z satisfying $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$, where arg denotes the principal argument. Then the set of all values of $|z-3+2i|$ for $z \in S$ is

(A) $(2\sqrt{2}, 4\sqrt{2}]$
(B) $[2\sqrt{2}, 4\sqrt{2}]$
(C) $(2\sqrt{2}, 4\sqrt{2})$
(D) $[2\sqrt{5}, 4\sqrt{2}]$

Only one option is correct.

हिन्दी

माना कि S उन सभी सम्मिश्र संख्याओं z का समुच्चय है जो $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$ को संतुष्ट करती हैं, जहाँ arg मुख्य कोणांक (principal argument) को दर्शाता है। तब $z \in S$ के लिए $|z-3+2i|$ के सभी मानों का समुच्चय है

(A) $(2\sqrt{2}, 4\sqrt{2}]$
(B) $[2\sqrt{2}, 4\sqrt{2}]$
(C) $(2\sqrt{2}, 4\sqrt{2})$
(D) $[2\sqrt{5}, 4\sqrt{2}]$

केवल एक विकल्प सही है।

Solution

For $z \neq \pm 1$, $\arg\left(\frac{z-1}{z+1}\right)$ is the angle through which the vector from -1 to z must turn to reach the vector from 1 to z, that is, the angle subtended at z by the segment from -1 to 1, taken with sign.

For $z = iy$, $y > 0$: $\arg(iy-1) - \arg(iy+1) = (\pi - \tan^{-1} y) - \tan^{-1} y = \pi - 2\tan^{-1} y \in (0, \pi)$. So the argument is positive above the real axis, and the locus is an arc in the upper half-plane.

An arc on which a chord of length 2 subtends $\frac{\pi}{4}$ belongs to a circle in which the chord subtends $\frac{\pi}{2}$ at the centre. So $2 = 2r\sin\frac{\pi}{4}$, $r = \sqrt{2}$, and the centre is at distance $\sqrt{r^2-1} = 1$ from the chord.

Since the inscribed angle $\frac{\pi}{4}$ is acute, the arc is the major arc and the centre is on the same side as the arc: $c = i$. Check: $z = i(1+\sqrt{2})$ gives $\pi - 2\tan^{-1}(1+\sqrt{2}) = \pi - \frac{3\pi}{4} = \frac{\pi}{4}$.

Hence $S = \{z : |z-i| = \sqrt{2},\ \mathrm{Im}\, z > 0\}$; the points ± 1 are not in S because the quotient is undefined at -1 and is 0 at 1.

Let $p = 3-2i$. Then $|p-c| = |3-3i| = 3\sqrt{2} > \sqrt{2}$, so p is outside the circle. By the triangle inequality, for every z on the circle, $3\sqrt{2} - \sqrt{2} \le |z-p| \le 3\sqrt{2} + \sqrt{2}$, with equality at the two ends of the diameter through p.

The far end is $c + \sqrt{2}\cdot\frac{c-p}{|c-p|} = i + \sqrt{2}\cdot\frac{-3+3i}{3\sqrt{2}} = -1+2i$, which has Im $= 2 > 0$, so it lies on the arc and $|z-p| = 4\sqrt{2}$ is attained.

The near end is $c + \sqrt{2}\cdot\frac{p-c}{|p-c|} = i + (1-i) = 1$, which is excluded. So $2\sqrt{2}$ is not attained, but points of the arc close to 1 give values as close to $2\sqrt{2}$ as we like.

At the other end point -1 (also excluded) the value would be $|-4+2i| = 2\sqrt{5} < 4\sqrt{2}$, which does not affect the range.

The arc is connected and $|z-p|$ is continuous, so every value between the infimum and the maximum occurs. The set of values is $(2\sqrt{2}, 4\sqrt{2}]$.

Correct Option: (A)

Question 03

English

Let $\omega = \dfrac{-1+i\sqrt{3}}{2}$, where $i = \sqrt{-1}$, and let $S = \displaystyle\sum_{k=0}^{9} \binom{9}{k} \omega^{k^2}$. Then S is equal to

(A) $1 + 171\sqrt{3}\,i$
(B) $-1 + 171\sqrt{3}\,i$
(C) $-1 - 171\sqrt{3}\,i$
(D) $170 + 171\sqrt{3}\,i$

Only one option is correct.

हिन्दी

माना कि $\omega = \dfrac{-1+i\sqrt{3}}{2}$ है, जहाँ $i = \sqrt{-1}$ है, तथा $S = \displaystyle\sum_{k=0}^{9} \binom{9}{k} \omega^{k^2}$ है। तब S बराबर है

(A) $1 + 171\sqrt{3}\,i$
(B) $-1 + 171\sqrt{3}\,i$
(C) $-1 - 171\sqrt{3}\,i$
(D) $170 + 171\sqrt{3}\,i$

केवल एक विकल्प सही है।

Solution

Since $\omega^3 = 1$, ω^{k^2} depends only on k^2 modulo 3.

If $3 \mid k$ then $k^2 \equiv 0$; if $k \equiv \pm 1 \pmod 3$ then $k^2 \equiv 1$. So $\omega^{k^2} = 1$ when $3 \mid k$ and $\omega^{k^2} = \omega$ otherwise.

Let $A = \displaystyle\sum_{3|k} \binom{9}{k}$. The remaining coefficients add up to $2^9 - A$, so $S = A + \omega(2^9 - A)$.

Multisection: $1 + \omega^j + \omega^{2j}$ is 3 if $3 \mid j$ and 0 otherwise. Hence $(1+1)^9 + (1+\omega)^9 + (1+\omega^2)^9 = \displaystyle\sum_{k} \binom{9}{k} \left(1 + \omega^k + \omega^{2k}\right) = 3A$.

$1 + \omega = \dfrac{1+i\sqrt{3}}{2} = e^{i\pi/3}$, a primitive sixth root of unity, so $(1+\omega)^9 = e^{3i\pi} = -1$; similarly $(1+\omega^2)^9 = e^{-3i\pi} = -1$.

$3A = 512 - 1 - 1 = 510$, so $A = 170$ and $2^9 - A = 342$. (Check: $\binom{9}{0} + \binom{9}{3} + \binom{9}{6} + \binom{9}{9} = 1+84+84+1 = 170$.)

$$S = 170 + 342\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = 170 - 171 + 171\sqrt{3}\,i = -1 + 171\sqrt{3}\,i.$$

Correct Option: (B)

Question 04

English

Consider the four lines in the Argand plane

$L_1 : (1-2i)z + (1+2i)\bar{z} + 4 = 0$, $L_2 : (2-4i)z + (2+4i)\bar{z} - 12 = 0$,

$L_3 : (3+i)z + (3-i)\bar{z} + 7 = 0$, $L_4 : (6+2i)z + (6-2i)\bar{z} - 14 = 0$.

Then the area of the parallelogram bounded by L_1, L_2, L_3 and L_4 is

(A) 10
(B) 12
(C) 20
(D) 5

Only one option is correct.

हिन्दी

आर्गण्ड तल में चार रेखाओं पर विचार कीजिए

$L_1 : (1-2i)z + (1+2i)\bar{z} + 4 = 0$, $L_2 : (2-4i)z + (2+4i)\bar{z} - 12 = 0$,

$L_3 : (3+i)z + (3-i)\bar{z} + 7 = 0$, $L_4 : (6+2i)z + (6-2i)\bar{z} - 14 = 0$.

तब L_1, L_2, L_3 तथा L_4 द्वारा परिबद्ध समान्तर चतुर्भुज का क्षेत्रफल है

(A) 10
(B) 12
(C) 20
(D) 5

केवल एक विकल्प सही है।

Solution

For $z = x + iy$ and $a = a_1 + ia_2$, $z\bar{a} + \bar{z}a = 2\,\mathrm{Re}(z\bar{a}) = 2(a_1x + a_2y)$. So $z\bar{a} + \bar{z}a + b = 0$ is the line $2a_1x + 2a_2y + b = 0$ with normal along a.

L_1: $\bar{a} = 1 - 2i$, $a = 1 + 2i$, $b = 4$: $2x + 4y + 4 = 0$, i.e. $x + 2y + 2 = 0$.

L_2: $\bar{a} = 2 - 4i$, $a = 2 + 4i$, $b = -12$: $4x + 8y - 12 = 0$, i.e. $x + 2y - 3 = 0$. Same normal, so $L_1 \parallel L_2$; the coefficient must be normalised before comparing constants.

Distance between them: $d_1 = \dfrac{|2 - (-3)|}{\sqrt{1+4}} = \sqrt{5}$.

L_3: $a = 3 - i$, $b = 7$: $6x - 2y + 7 = 0$. L_4: $a = 6 - 2i$, $b = -14$: $12x - 4y - 14 = 0$, i.e. $6x - 2y - 7 = 0$.

So $L_3 \parallel L_4$ and $d_2 = \dfrac{14}{\sqrt{40}} = \dfrac{7}{\sqrt{10}}$.

The normals $(1, 2)$ and $(3, -1)$ make an angle ϕ with $\cos\phi = \dfrac{3-2}{\sqrt{5}\sqrt{10}} = \dfrac{1}{5\sqrt{2}}$ and $\sin\phi = \dfrac{|1 \cdot (-1) - 2 \cdot 3|}{\sqrt{5}\sqrt{10}} = \dfrac{7}{5\sqrt{2}}$; the angle between the lines is the same.

A parallelogram whose opposite sides are at distances d_1 and d_2 and whose sides meet at angle ϕ has sides $\dfrac{d_2}{\sin\phi}$ and $\dfrac{d_1}{\sin\phi}$ and area $\dfrac{d_1}{\sin\phi} \cdot \dfrac{d_2}{\sin\phi} \cdot \sin\phi = \dfrac{d_1 d_2}{\sin\phi}$.

$$\text{Area} = \sqrt{5} \cdot \frac{7}{\sqrt{10}} \cdot \frac{5\sqrt{2}}{7} = \frac{5\sqrt{10}}{\sqrt{10}} = 5.$$

(Using $\dfrac{|b|}{|a|}$ instead of $\dfrac{|b|}{2|a|}$ for the distances gives 20; not normalising L_2 and L_4 gives 12.)

Correct Option: (D)

Question 05

English

Let S be the set of all non-zero complex numbers z satisfying $z + \frac{9}{\bar{z}} = 6 + 8i$, where $i = \sqrt{-1}$. Then the sum of the values of $|z|^2$ over all $z \in S$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी शून्येतर सम्मिश्र संख्याओं z का समुच्चय है जो $z + \frac{9}{\bar{z}} = 6 + 8i$ को संतुष्ट करती हैं, जहाँ $i = \sqrt{-1}$ है। तब सभी $z \in S$ पर $|z|^2$ के मानों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Since $z \neq 0$, $\frac{1}{\bar{z}} = \frac{z}{z\bar{z}} = \frac{z}{|z|^2}$. So the equation becomes $z + \frac{9z}{|z|^2} = 6 + 8i$, i.e. $z\left(1 + \frac{9}{|z|^2}\right) = 6 + 8i$.

The factor $1 + \frac{9}{|z|^2}$ is a positive real number, so z has the same argument as $6 + 8i$: $z = r \cdot \frac{6 + 8i}{10} = r \cdot \frac{3 + 4i}{5}$ with $r = |z| > 0$.

Taking moduli of both sides: $r\left(1 + \frac{9}{r^2}\right) = |6 + 8i| = 10$, so $r^2 - 10r + 9 = 0$, $(r - 1)(r - 9) = 0$, $r = 1$ or $r = 9$.

$r = 1$: $z = \frac{3 + 4i}{5}$; check: $\bar{z} = \frac{3 - 4i}{5}$, $\frac{9}{\bar{z}} = 9 \cdot \frac{3 + 4i}{5}$, sum $= 10 \cdot \frac{3 + 4i}{5} = 6 + 8i$.

$r = 9$: $z = \frac{27 + 36i}{5}$; check: $\frac{9}{\bar{z}} = \frac{9z}{81} = \frac{3 + 4i}{5}$, sum $= \frac{30 + 40i}{5} = 6 + 8i$.

So S has exactly two elements, with $|z|^2 = 1$ and 81. The required sum is 82.

Final Answer: 82

Question 06

English

Let S be the set of all complex numbers z satisfying $|z-4-2i|=2$, where $i=\sqrt{-1}$. Then the minimum value of $\operatorname{Re}\left(\frac{\bar{z}}{z}\right)+\operatorname{Im}\left(\frac{\bar{z}}{z}\right)$ over $z \in S$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी सम्मिश्र संख्याओं z का समुच्चय है जो $|z-4-2i|=2$ को संतुष्ट करती हैं, जहाँ $i=\sqrt{-1}$ है। तब $z \in S$ पर $\operatorname{Re}\left(\frac{\bar{z}}{z}\right)+\operatorname{Im}\left(\frac{\bar{z}}{z}\right)$ का न्यूनतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $z \neq 0$ write $z=re^{i\theta}$. Then $\bar{z}=re^{-i\theta}$ and $\dfrac{\bar{z}}{z}=e^{-2i\theta}=\cos 2\theta - i\sin 2\theta$.

So the expression equals $\cos 2\theta - \sin 2\theta = \sqrt{2}\cos\left(2\theta+\dfrac{\pi}{4}\right)$; it depends only on $\arg z$.

The circle has centre $4+2i$ and radius 2; its distance from the real axis is 2, so it touches the real axis at $z=4$. The origin is outside ($|4+2i|=2\sqrt{5}>2$), so $z \neq 0$ on S.

Let $\phi=\arg(4+2i)$, so $\tan\phi=\dfrac{1}{2}$. The two tangents from the origin are symmetric about the line through the centre; one is the real axis (angle 0), so the other makes angle 2ϕ. Hence $\arg z$ takes every value in $[0,2\phi]$ on S.

$\tan 2\phi = \dfrac{2\cdot\frac{1}{2}}{1-\frac{1}{4}} = \dfrac{4}{3}$, so $2\phi \in \left(0,\dfrac{\pi}{2}\right)$ and $t=2\theta$ runs over $[0,4\phi]$, where $4\phi<\pi$.

$\cos 4\phi = \dfrac{1-\tan^2 2\phi}{1+\tan^2 2\phi} = \dfrac{1-\frac{16}{9}}{1+\frac{16}{9}} = -\dfrac{7}{25}$ and $\sin 4\phi = \dfrac{2\tan 2\phi}{1+\tan^2 2\phi} = \dfrac{8/3}{25/9} = \dfrac{24}{25}$.

Since $\cos 4\phi = -\dfrac{7}{25} > -\dfrac{1}{\sqrt{2}} = \cos\dfrac{3\pi}{4}$ and both angles lie in $(0,\pi)$, $4\phi < \dfrac{3\pi}{4}$. Hence for $t \in [0,4\phi]$, $t+\dfrac{\pi}{4} \in \left[\dfrac{\pi}{4},\pi\right)$, where cos is decreasing; so the least value occurs at $t=4\phi$, the largest argument.

Minimum at $t=4\phi$: $\cos 4\phi - \sin 4\phi = -\dfrac{7}{25}-\dfrac{24}{25} = -\dfrac{31}{25} = -1.24$, attained at the point of contact $z=\dfrac{12+16i}{5}$ of the second tangent.

Final Answer: -1.24

Question 07

English

Let O be the origin of the Argand plane and let $OBCD$ be a square whose vertices O, B, C, D are in the anticlockwise order. The vertex B lies on the circle $|z-3-i| = \sqrt{5}$ and the vertex D lies on the line $z + \bar{z} + 4 = 0$, where $i = \sqrt{-1}$. If C represents the complex number c, then the product of all possible values of $\mathrm{Re}(c)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि O आर्गण्ड तल का मूल बिन्दु है तथा $OBCD$ एक वर्ग है जिसके शीर्ष O, B, C, D वामावर्त (anticlockwise) क्रम में हैं। शीर्ष B वृत्त $|z-3-i| = \sqrt{5}$ पर स्थित है तथा शीर्ष D रेखा $z + \bar{z} + 4 = 0$ पर स्थित है, जहाँ $i = \sqrt{-1}$ है। यदि C सम्मिश्र संख्या c को निरूपित करता है, तो $\mathrm{Re}(c)$ के सभी सम्भव मानों का गुणनफल _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let B, C, D represent b, c, d. In the square $OBCD$ taken anticlockwise, $\overrightarrow{OD}$ is $\overrightarrow{OB}$ turned through $\frac{\pi}{2}$ anticlockwise, so $d = e^{i\pi/2}b = ib$, with $|d| = |b|$.

The fourth vertex satisfies $c = b + d$ (the diagonals of a square bisect each other), so $c = (1+i)b$.

Since $|b-(3+i)| = \sqrt{5}$, multiplying by i (a rotation, which keeps moduli): $|ib - i(3+i)| = \sqrt{5}$, i.e. $|d-(-1+3i)| = \sqrt{5}$. So D lies on the circle with centre $-1+3i$ and radius $\sqrt{5}$.

For $z = x+iy$, $z+\bar{z} = 2x$, so the line is $x = -2$. Hence $d = -2+iy$ with $(-2+1)^2 + (y-3)^2 = 5$, $(y-3)^2 = 4$, $y = 1$ or $y = 5$.

Case $d = -2+i$: $b = -id = -i(-2+i) = 1+2i$; check $|1+2i-3-i| = |-2+i| = \sqrt{5}$. Then $c = b+d = -1+3i$.

Case $d = -2+5i$: $b = -id = 5+2i$; check $|2+i| = \sqrt{5}$. Then $c = 3+7i$.

Both give genuine squares ($b \neq 0$). The possible values of $\mathrm{Re}(c)$ are -1 and 3, with product -3.

Final Answer: -3

Question 08

English

Let $z = \dfrac{\sqrt{6}}{4}\left(\sqrt{3}+i\right)$, where $i = \sqrt{-1}$. Then the value of $\displaystyle\sum_{k=0}^{4} z^k\,\bar{z}^{\,4-k}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $z = \dfrac{\sqrt{6}}{4}\left(\sqrt{3}+i\right)$ है, जहाँ $i = \sqrt{-1}$ है। तब $\displaystyle\sum_{k=0}^{4} z^k\,\bar{z}^{\,4-k}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$|z|^2 = \dfrac{6}{16}(3+1) = \dfrac{3}{2}$, and $\sqrt{3}+i = 2e^{i\pi/6}$, so $z = re^{i\pi/6}$ with $r = \sqrt{3/2}$, and $\bar{z} = re^{-i\pi/6}$.

Each term is $z^k\bar{z}^{4-k} = r^4 e^{i(2k-4)\pi/6}$, so the sum is $r^4\left(e^{-2i\pi/3} + e^{-i\pi/3} + 1 + e^{i\pi/3} + e^{2i\pi/3}\right)$.

Equivalently, since $z \neq \bar{z}$, the five terms form a GP with ratio $\dfrac{z}{\bar{z}}$, and $\displaystyle\sum_{k=0}^{4} z^k\bar{z}^{4-k} = \frac{z^5 - \bar{z}^5}{z - \bar{z}}$.

By De Moivre's theorem, $z^5 = r^5 e^{5i\pi/6}$, so $z^5 - \bar{z}^5 = 2ir^5 \sin\dfrac{5\pi}{6} = ir^5$; and $z - \bar{z} = 2ir\sin\dfrac{\pi}{6} = ir$.

Hence the sum is $\dfrac{ir^5}{ir} = r^4 = \dfrac{9}{4}$.

Check with the bracket: $1 + 2\cos\dfrac{\pi}{3} + 2\cos\dfrac{2\pi}{3} = 1 + 1 - 1 = 1$, so the sum is $r^4 \cdot 1 = 2.25$.

Final Answer: 2.25

Question 09

English

Let z_1, z_2, z_3 be distinct complex numbers with $|z_1| = |z_2| = |z_3| = 1$ and $z_1 z_2 z_3 = i$, where $i = \sqrt{-1}$. The orthocentre of the triangle with vertices z_1, z_2, z_3 is the point $\frac{12}{25} + \frac{16}{25}i$. Then the value of $\mathrm{Re}\,(z_1 z_2 + z_2 z_3 + z_3 z_1)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि z_1, z_2, z_3 भिन्न सम्मिश्र संख्याएँ हैं जिनके लिए $|z_1| = |z_2| = |z_3| = 1$ तथा $z_1 z_2 z_3 = i$ है, जहाँ $i = \sqrt{-1}$ है। शीर्षों z_1, z_2, z_3 वाले त्रिभुज का लम्बकेन्द्र बिन्दु $\frac{12}{25} + \frac{16}{25}i$ है। तब $\mathrm{Re}\,(z_1 z_2 + z_2 z_3 + z_3 z_1)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The vertices lie on $|z| = 1$, so the circumcentre O is the origin. The centroid is $g = \dfrac{z_1 + z_2 + z_3}{3}$, and on the Euler line the orthocentre H satisfies $\overrightarrow{OH} = 3\overrightarrow{OG}$.

Hence $h = z_1 + z_2 + z_3 = \dfrac{12}{25} + \dfrac{16}{25}i$. (Direct check: $(h - z_1)$ is $z_2 + z_3$, and $\dfrac{z_2 + z_3}{z_2 - z_3}$ is purely imaginary for $|z_2| = |z_3|$, so H lies on the altitude from z_1.)

For $|z_k| = 1$, $\bar{z}_k = \dfrac{1}{z_k}$. Taking the conjugate of h: $\bar{h} = \dfrac{1}{z_1} + \dfrac{1}{z_2} + \dfrac{1}{z_3} = \dfrac{z_2 z_3 + z_3 z_1 + z_1 z_2}{z_1 z_2 z_3}$.

So $z_1 z_2 + z_2 z_3 + z_3 z_1 = z_1 z_2 z_3\, \bar{h} = i\left(\dfrac{12}{25} - \dfrac{16}{25}i\right) = \dfrac{16}{25} + \dfrac{12}{25}i$.

Existence: z_1, z_2, z_3 are the roots of $t^3 - ht^2 + \dfrac{16 + 12i}{25}t - i = 0$, whose three roots are distinct and of modulus 1 (approximately $0.690 + 0.724i$, $-0.474 + 0.881i$, $0.264 - 0.964i$), so the data are consistent.

Required real part $= \dfrac{16}{25} = 0.64$.

Final Answer: 0.64

Question 10

English

Let S be the set of all complex numbers z satisfying $|z-8| = 3|z|$. Let M and m be the maximum and the minimum values of $|z-14-8i|$ over $z \in S$, where $i = \sqrt{-1}$. Then the value of $M^2 + m^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी सम्मिश्र संख्याओं z का समुच्चय है जो $|z-8| = 3|z|$ को संतुष्ट करती हैं। माना कि $z \in S$ पर $|z-14-8i|$ के अधिकतम तथा न्यूनतम मान क्रमशः M तथा m हैं, जहाँ $i = \sqrt{-1}$ है। तब M^2+m^2 का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Put $z = x + iy$. Then $|z-8| = 3|z|$ gives $(x-8)^2 + y^2 = 9(x^2+y^2)$, i.e. $8x^2 + 8y^2 + 16x - 64 = 0$, i.e. $x^2 + y^2 + 2x - 8 = 0$.

So S is the circle $(x+1)^2 + y^2 = 9$: centre $c = -1$, radius 3. (Check: $z = 2$ gives $6 = 3 \cdot 2$; $z = -4$ gives $12 = 3 \cdot 4$.)

Let $p = 14 + 8i$. Then $|p - c| = |15 + 8i| = \sqrt{225 + 64} = 17$.

For z on the circle, $|z-p| \le |z-c| + |c-p| = 3 + 17 = 20$ and $|z-p| \ge |c-p| - |z-c| = 17 - 3 = 14$.

Equality holds at the two points where the line through c and p meets the circle, so $M = 20$ and $m = 14$.

$M^2 + m^2 = 400 + 196 = 596$.

Final Answer: 596

Question 11

English

Let $S = \{z \in \mathbb{C} : |z - 3 - 4i| = 2\}$, where $i = \sqrt{-1}$. Let M and m be the maximum and the minimum values of $|z^2 + 7 - 24i|$ over $z \in S$. Then the value of $m - M$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $S = \{z \in \mathbb{C} : |z - 3 - 4i| = 2\}$ है, जहाँ $i = \sqrt{-1}$ है। माना कि $z \in S$ पर $|z^2 + 7 - 24i|$ के अधिकतम तथा न्यूनतम मान क्रमशः M तथा m हैं। तब $m - M$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Note that $(3 + 4i)^2 = 9 - 16 + 24i = -7 + 24i$. Hence $z^2 + 7 - 24i = z^2 - (3 + 4i)^2 = (z - (3 + 4i))(z + (3 + 4i))$.

For $z \in S$, $|z - (3 + 4i)| = 2$, so $|z^2 + 7 - 24i| = 2\,|z - (-3 - 4i)|$.

The point $q = -3 - 4i$ is at distance $|(3 + 4i) - (-3 - 4i)| = |6 + 8i| = 10$ from the centre of S, which exceeds the radius 2.

By the triangle inequality, $10 - 2 \le |z - q| \le 10 + 2$ for $z \in S$, with equality at $z = 3 + 4i \mp \dfrac{2}{10}(6 + 8i)$, i.e. at $z = 1.8 + 2.4i$ (minimum) and $z = 4.2 + 5.6i$ (maximum).

Therefore $m = 2 \cdot 8 = 16$ and $M = 2 \cdot 12 = 24$. Check at $z = 4.2 + 5.6i$: $z^2 = -13.72 + 47.04i$, $z^2 + 7 - 24i = -6.72 + 23.04i$, modulus $\sqrt{45.1584 + 530.8416} = 24$.

$m - M = 16 - 24 = -8$.

Final Answer: -8

Question 12

English

Let T be the set of all complex numbers $z = x + iy$, with x and y integers, $z \neq \pm 2$ and $i = \sqrt{-1}$, satisfying $\frac{\pi}{4} \leq \arg\left(\frac{z-2}{z+2}\right) \leq \frac{\pi}{2}$, where arg denotes the principal argument. Then the number of elements in T is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि T उन सभी सम्मिश्र संख्याओं $z = x + iy$ का समुच्चय है, जहाँ x तथा y पूर्णांक हैं, $z \neq \pm 2$ है और $i = \sqrt{-1}$ है, जो $\frac{\pi}{4} \leq \arg\left(\frac{z-2}{z+2}\right) \leq \frac{\pi}{2}$ को संतुष्ट करती हैं, जहाँ arg मुख्य कोणांक को दर्शाता है। तब T में अवयवों की संख्या _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $z \neq \pm 2$, $\frac{z-2}{z+2} = \frac{(z-2)(\bar{z}+2)}{|z+2|^2}$ and $(z-2)(\bar{z}+2) = (x^2+y^2-4) + 4iy$.

So the argument lies in $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$ exactly when $4y > 0$, $x^2+y^2-4 \geq 0$ and $4y \geq x^2+y^2-4$ (imaginary part at least the real part, both on the correct side).

Geometrically: $x^2 + y^2 - 4 = 0$ is the circle on the segment from -2 to 2 as diameter (angle $\frac{\pi}{2}$), and $x^2 + y^2 - 4y - 4 = 0$, i.e. $x^2 + (y-2)^2 = 8$, is the circle through ± 2 whose upper major arc sees the segment at $\frac{\pi}{4}$. The region lies between the two arcs, boundaries included.

Hence T consists of the lattice points with $y \geq 1$, $x^2 + y^2 \geq 4$ and $x^2 + (y-2)^2 \leq 8$.

$y = 1$: need $x^2 \geq 3$ and $x^2 \leq 7$, so $x = \pm 2$: 2 points.

$y = 2$: need $x^2 \geq 0$ and $x^2 \leq 8$: $x = -2, \ldots, 2$: 5 points (including $2i$ on the semicircle).

$y = 3$: $x^2 \leq 7$: $x = -2, \ldots, 2$: 5 points. $y = 4$: $x^2 \leq 4$: $x = -2, \ldots, 2$: 5 points (including $\pm 2 + 4i$ on the larger circle).

$y \geq 5$: $(y-2)^2 \geq 9 > 8$, no points. Points with $y \leq 0$ give an argument outside the interval.

Total $= 2 + 5 + 5 + 5 = 17$.

Final Answer: 17

Question 13

English

Let P_0 and P_1 be the points representing the complex numbers 0 and 4 in the Argand plane. For each integer $n \geq 0$, let P_{n+2} be the point such that the triangle $P_nP_{n+1}P_{n+2}$ is isosceles with the right angle at P_{n+2}, and P_{n+2} lies to the right of the directed line from P_n to P_{n+1}. As $n \to \infty$, the points P_n approach the point representing $\alpha + i\beta$, where α, β are real and $i = \sqrt{-1}$. Then the value of $\alpha\beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि P_0 तथा P_1 आर्गण्ड तल में सम्मिश्र संख्याओं 0 तथा 4 को निरूपित करने वाले बिन्दु हैं। प्रत्येक पूर्णांक $n \geq 0$ के लिए माना कि P_{n+2} वह बिन्दु है जिसके लिए त्रिभुज $P_nP_{n+1}P_{n+2}$ समद्विबाहु है तथा उसका समकोण P_{n+2} पर है, और P_{n+2}, P_n से P_{n+1} की ओर निर्दिष्ट रेखा के दाईं ओर स्थित है। जब $n \to \infty$, तब बिन्दु P_n उस बिन्दु की ओर अग्रसर होते हैं जो $\alpha + i\beta$ को निरूपित करता है, जहाँ α, β वास्तविक हैं तथा $i = \sqrt{-1}$ है। तब $\alpha\beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let p_n be the complex number of P_n. In a right isosceles triangle with the right angle at P_{n+2}, the side P_nP_{n+2} has length $\frac{1}{\sqrt{2}}P_nP_{n+1}$ and makes an angle $\frac{\pi}{4}$ with P_nP_{n+1}.

Since P_{n+2} is to the right of the directed line $P_n \to P_{n+1}$, the turn from $\overrightarrow{P_nP_{n+1}}$ to $\overrightarrow{P_nP_{n+2}}$ is clockwise: $p_{n+2} - p_n = \frac{1}{\sqrt{2}}e^{-i\pi/4}(p_{n+1} - p_n) = \frac{1-i}{2}(p_{n+1} - p_n)$.

Check for $n = 0$: $p_2 = \frac{1-i}{2} \cdot 4 = 2 - 2i$, below the real axis (to the right of $0 \to 4$), and $|p_2 - 0| = |p_2 - 4| = 2\sqrt{2}$, $(p_2 - 0) \perp (p_2 - 4)$.

So $p_{n+2} = c\,p_n + (1-c)\,p_{n+1}$ with $c = \frac{1+i}{2}$, $1 - c = \frac{1-i}{2}$.

Then $p_{n+2} + c\,p_{n+1} = c\,p_n + (1-c)p_{n+1} + c\,p_{n+1} = p_{n+1} + c\,p_n$. Hence $p_{n+1} + c\,p_n$ is the same for all n, equal to $p_1 + c\,p_0 = 4$.

Convergence: the characteristic equation $\lambda^2 - (1-c)\lambda - c = 0$ has roots 1 and $-c$, with $|-c| = \frac{1}{\sqrt{2}} < 1$; so $p_n = A + B(-c)^n \to A$.

Letting $n \to \infty$ in $p_{n+1} + c\,p_n = 4$: $L(1+c) = 4$, $L = \frac{4}{(3+i)/2} = \frac{8}{3+i} = \frac{8(3-i)}{10} = 2.4 - 0.8i$.

Numerical check: $p_3 = 2$, $p_4 = 3 - i$, $p_5 = 2 - i$, $p_6 = 2.5 - 0.5i$, and the points spiral in to $2.4 - 0.8i$.

$\alpha = 2.4$, $\beta = -0.8$, so $\alpha\beta = -1.92$.

Final Answer: -1.92

Question 14

English

Let T be the set of all real numbers t for which the principal argument of $(1+it)^3$ is $\dfrac{3\pi}{4}$, where $i=\sqrt{-1}$. Then the sum of the values of $\dfrac{2t}{1+t^2}$ over all $t \in T$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि T उन सभी वास्तविक संख्याओं t का समुच्चय है जिनके लिए $(1+it)^3$ का मुख्य कोणांक $\dfrac{3\pi}{4}$ है, जहाँ $i=\sqrt{-1}$ है। तब सभी $t \in T$ पर $\dfrac{2t}{1+t^2}$ के मानों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Every real t is $\tan\theta$ for a unique $\theta \in \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$, and $1+i\tan\theta = \sec\theta(\cos\theta + i\sin\theta)$ with $\sec\theta > 0$.

By De Moivre's theorem, $(1+it)^3 = \sec^3\theta(\cos 3\theta + i\sin 3\theta)$, so an argument of $(1+it)^3$ is 3θ, where $3\theta \in \left(-\dfrac{3\pi}{2}, \dfrac{3\pi}{2}\right)$.

The principal argument is $\dfrac{3\pi}{4}$ exactly when $3\theta = \dfrac{3\pi}{4} + 2k\pi$ for an integer k. In the interval this allows $3\theta = \dfrac{3\pi}{4}$ $(k=0)$ and $3\theta = -\dfrac{5\pi}{4}$ $(k=-1)$.

So $\theta = \dfrac{\pi}{4}$, $t=1$, or $\theta = -\dfrac{5\pi}{12}$, $t = -\tan\dfrac{5\pi}{12} = -(2+\sqrt{3})$.

Algebraic check: $(1+it)^3 = (1-3t^2) + i(3t-t^3)$. Argument $\dfrac{3\pi}{4}$ needs $3t - t^3 = -(1-3t^2) > 0$, i.e. $t^3 + 3t^2 - 3t - 1 = 0$, $(t-1)(t^2+4t+1) = 0$, $t = 1, -2 \pm \sqrt{3}$. For $t = -2+\sqrt{3}$, $1-3t^2 > 0$, so the argument is $-\dfrac{\pi}{4}$ and this root is rejected; $t=1$ and $t = -2-\sqrt{3}$ give real part < 0 and imaginary part > 0, as required.

Using the multiple-angle formula, $\dfrac{2t}{1+t^2} = \dfrac{2\tan\theta}{1+\tan^2\theta} = \sin 2\theta$.

$t=1$: $\sin\dfrac{\pi}{2} = 1$. $t = -2-\sqrt{3}$: $\sin\left(-\dfrac{5\pi}{6}\right) = -\dfrac{1}{2}$ (directly: $\dfrac{2(-2-\sqrt{3})}{8+4\sqrt{3}} = -\dfrac{1}{2}$).

Sum $= 1 - \dfrac{1}{2} = 0.5$.

Final Answer: 0.5

Question 15

English

For each positive integer n, let $S_n = \sum_{k=1}^{n} 2^k \cos\frac{k\pi}{3}$. Then the greatest value of S_n for $1 \le n \le 10$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

प्रत्येक धनात्मक पूर्णांक n के लिए माना कि $S_n = \sum_{k=1}^{n} 2^k \cos\frac{k\pi}{3}$ है। तब $1 \le n \le 10$ के लिए S_n का महत्तम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $u = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) = 1 + i\sqrt{3}$. By De Moivre's theorem, $u^k = 2^k\left(\cos\frac{k\pi}{3} + i\sin\frac{k\pi}{3}\right)$, so $S_n = \mathrm{Re}\sum_{k=1}^{n} u^k$.

The angles $\frac{k\pi}{3}$ are in AP, so the series is a GP in u: $\sum_{k=1}^{n} u^k = \frac{u(u^n-1)}{u-1}$ $(u \ne 1)$.

$u - 1 = i\sqrt{3}$, so $\frac{u}{u-1} = \frac{1+i\sqrt{3}}{i\sqrt{3}} = \frac{(1+i\sqrt{3})(-i)}{\sqrt{3}} = 1 - \frac{i}{\sqrt{3}}$.

With $u^n = 2^n(\cos n\pi/3 + i\sin n\pi/3)$: $\mathrm{Re}\left[\left(1 - \frac{i}{\sqrt{3}}\right)(u^n - 1)\right] = 2^n\cos\frac{n\pi}{3} + \frac{2^n}{\sqrt{3}}\sin\frac{n\pi}{3} - 1$.

$\cos A + \frac{1}{\sqrt{3}}\sin A = \frac{2}{\sqrt{3}}\left(\sin\frac{\pi}{3}\cos A + \cos\frac{\pi}{3}\sin A\right) = \frac{2}{\sqrt{3}}\sin\left(A + \frac{\pi}{3}\right)$, so $S_n = \frac{2^{n+1}}{\sqrt{3}}\sin\frac{(n+1)\pi}{3} - 1$.

$n \equiv 0$ or $1 \pmod 6$: $\sin = \frac{\sqrt{3}}{2}$, $S_n = 2^n - 1$. $n \equiv 2$ or 5: $\sin = 0$, $S_n = -1$. $n \equiv 3$ or 4: $\sin = -\frac{\sqrt{3}}{2}$, $S_n = -2^n - 1$.

For $n = 1, \ldots, 10$: $S = 1, -1, -9, -17, -1, 63, 127, -1, -513, -1025$. (Check: $S_1 = 2 \cdot \frac{1}{2} = 1$, $S_2 = 1 - 2 = -1$.)

The greatest value is $S_7 = 2^7 - 1 = 127$.

Final Answer: 127

Question 16

English

Let $\omega = \dfrac{-1+i\sqrt{3}}{2}$, where $i = \sqrt{-1}$, and let a, b, c be real numbers such that $a+b+c = \dfrac{1}{2}$ and $|a+b\omega+c\omega^2| = \sqrt{7}$. Then the value of the determinant $\begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $\omega = \dfrac{-1+i\sqrt{3}}{2}$ है, जहाँ $i = \sqrt{-1}$ है, तथा a, b, c ऐसी वास्तविक संख्याएँ हैं कि $a+b+c = \dfrac{1}{2}$ तथा $|a+b\omega+c\omega^2| = \sqrt{7}$ है। तब सारणिक $\begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Expanding along the first row: $D = a(a^2-bc) - b(ca-b^2) + c(c^2-ab) = a^3+b^3+c^3-3abc$.

With $1+\omega+\omega^2 = 0$ and $\omega^3 = 1$, $(a+b\omega+c\omega^2)(a+b\omega^2+c\omega) = a^2+b^2+c^2-ab-bc-ca$, and multiplying by $a+b+c$ gives $a^3+b^3+c^3-3abc$.

(Equivalently, adding all columns to the first shows $a+b+c$ is a factor, and adding ω and ω^2 times the columns shows the other two factors.)

Since a, b, c are real and $\bar{\omega} = \omega^2$, $\overline{a+b\omega+c\omega^2} = a+b\omega^2+c\omega$. So $(a+b\omega+c\omega^2)(a+b\omega^2+c\omega) = |a+b\omega+c\omega^2|^2 = 7$.

$D = (a+b+c)\cdot 7 = \dfrac{7}{2} = 3.5$.

Such a, b, c exist, for example $a = \dfrac{3}{2}$, $b = \dfrac{1}{2}$, $c = -\dfrac{3}{2}$, for which $a+b\omega+c\omega^2 = 2+i\sqrt{3}$; the value does not depend on the choice.

Final Answer: 3.5

Question 17

English

The value of $\displaystyle\sum_{k=0}^{14} \binom{14}{k} \cos^2 \frac{k\pi}{3}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$\displaystyle\sum_{k=0}^{14} \binom{14}{k} \cos^2 \frac{k\pi}{3}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For integer k, $\cos \dfrac{k\pi}{3} \in \left\{\pm 1, \pm\dfrac{1}{2}\right\}$: it is ± 1 exactly when $3 \mid k$. So $\cos^2 \dfrac{k\pi}{3} = 1$ if $3 \mid k$ and $\dfrac{1}{4}$ otherwise.

Let $A = \displaystyle\sum_{3|k} \binom{14}{k}$. The sum equals $A + \dfrac{1}{4}\left(2^{14} - A\right) = \dfrac{3A}{4} + 2^{12}$.

Let $\omega = e^{2\pi i/3}$. Since $1 + \omega^k + \omega^{2k}$ is 3 for $3 \mid k$ and 0 otherwise, $3A = 2^{14} + (1+\omega)^{14} + (1+\omega^2)^{14}$.

$1 + \omega = e^{i\pi/3}$ and $1 + \omega^2 = e^{-i\pi/3}$ are sixth roots of unity, so $(1+\omega)^{14} + (1+\omega^2)^{14} = 2\cos\dfrac{14\pi}{3} = 2\cos\dfrac{2\pi}{3} = -1$.

$3A = 16383$, $A = 5461$.

Sum $= \dfrac{3 \cdot 5461}{4} + 4096 = 4095.75 + 4096 = 8191.75$.

(Check by $\cos^2 x = \dfrac{1 + \cos 2x}{2}$: the sum is $2^{13} + \dfrac{1}{2}\mathrm{Re}(1+\omega)^{14} = 8192 - \dfrac{1}{4} = 8191.75$.)

Final Answer: 8191.75

Question 18

English

For a real number b, let L_b denote the line $(3-4i)z+(3+4i)\bar{z}+b=0$ in the Argand plane, where $i=\sqrt{-1}$. Let b_0 be the greater of the two values of b for which L_b touches the circle $|z-1-2i|=2$, and let L_{b_0} touch this circle at the point z_0. Then the value of $\mathrm{Re}(z_0)-\mathrm{Im}(z_0)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

एक वास्तविक संख्या b के लिए माना कि L_b आर्गण्ड तल में रेखा $(3-4i)z+(3+4i)\bar{z}+b=0$ को दर्शाता है, जहाँ $i=\sqrt{-1}$ है। माना कि b के उन दो मानों में से, जिनके लिए L_b वृत्त $|z-1-2i|=2$ को स्पर्श करती है, बड़ा मान b_0 है, तथा L_{b_0} इस वृत्त को बिन्दु z_0 पर स्पर्श करती है। तब $\mathrm{Re}(z_0)-\mathrm{Im}(z_0)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

With $a=3+4i$, the line $z\bar{a}+\bar{z}a+b=0$ is $2\,\mathrm{Re}(z\bar{a})+b=0$, i.e. $6x+8y+b=0$, with normal along a.

The circle has centre $c=1+2i$ and radius 2. The distance of c from L_b is $\dfrac{|6+16+b|}{\sqrt{36+64}}=\dfrac{|22+b|}{10}$.

Tangency: $|22+b|=20$, so $b=-2$ or $b=-42$. The greater value is $b_0=-2$, giving $3x+4y=1$.

The point of contact is the foot of the perpendicular from c to the line. Along the unit normal $\dfrac{a}{|a|}=\dfrac{3+4i}{5}$, the value of $3x+4y$ at c is 11, larger than 1, so we move from c in the direction $-\dfrac{a}{|a|}$ through the radius 2.

$z_0=1+2i-\dfrac{2}{5}(3+4i)=-\dfrac{1}{5}+\dfrac{2}{5}i$. Check: $3\left(-\dfrac{1}{5}\right)+4\cdot\dfrac{2}{5}=1$ and $|z_0-c|=\left|-\dfrac{6}{5}-\dfrac{8}{5}i\right|=2$.

$\mathrm{Re}(z_0)-\mathrm{Im}(z_0)=-\dfrac{1}{5}-\dfrac{2}{5}=-0.6$.

Final Answer: -0.6

Question 19

English

Let C be the locus of the complex numbers z satisfying $|z| = 3|z-4|$. The ray $\arg z = \dfrac{\pi}{12}$ meets C at two points P and Q. If O is the origin, then the value of $OP \cdot OQ$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि C उन सम्मिश्र संख्याओं z का बिन्दुपथ है जो $|z| = 3|z-4|$ को संतुष्ट करती हैं। किरण $\arg z = \dfrac{\pi}{12}$, C को दो बिन्दुओं P तथा Q पर मिलती है। यदि O मूल बिन्दु है, तो $OP \cdot OQ$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

With $z = x+iy$: $x^2+y^2 = 9\left[(x-4)^2+y^2\right]$, i.e. $8x^2+8y^2-72x+144=0$, i.e. $x^2+y^2-9x+18=0$.

So C is the Apollonius circle $\left(x-\dfrac{9}{2}\right)^2+y^2=\dfrac{9}{4}$: centre $\dfrac{9}{2}$, radius $\dfrac{3}{2}$. (It meets the real axis at 3 and 6, which divide the segment from 0 to 4 internally and externally in the ratio $3:1$.)

Points of the ray $\arg z = \dfrac{\pi}{12}$ are $z = r\left(\cos\dfrac{\pi}{12}+i\sin\dfrac{\pi}{12}\right)$ with $r>0$, and $x^2+y^2=r^2$, $x = r\cos\dfrac{\pi}{12}$.

Substituting: $r^2 - 9r\cos\dfrac{\pi}{12}+18=0$.

The discriminant is $81\cos^2\dfrac{\pi}{12} - 72 = 81\cdot\dfrac{2+\sqrt{3}}{4} - 72 > 0$; the sum $9\cos\dfrac{\pi}{12}$ and the product 18 of the roots are positive, so both roots are positive and give the points P, Q on the ray. (Also $\sin\dfrac{\pi}{12} < \dfrac{1}{3} = \dfrac{3/2}{9/2}$, so the ray cuts the circle.)

Hence $OP\cdot OQ = r_1 r_2 = 18$, which is the power of O: $0^2+0^2-9\cdot 0+18=18$.

Final Answer: 18

Question 20

English

Let C be the circle $(z-1-2i)(\bar{z}-7+10i)+(\bar{z}-1+2i)(z-7-10i)=0$ in the Argand plane, where $i=\sqrt{-1}$. Suppose that C is also the locus of the points z satisfying $|z-\alpha|=k|z-5-6i|$ for a complex number α and a real number $k>1$. Then the value of $|\alpha|^2+k^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि C आर्गण्ड तल में वृत्त $(z-1-2i)(\bar{z}-7+10i)+(\bar{z}-1+2i)(z-7-10i)=0$ है, जहाँ $i=\sqrt{-1}$ है। मान लीजिए कि C, किसी सम्मिश्र संख्या α तथा वास्तविक संख्या $k>1$ के लिए $|z-\alpha|=k|z-5-6i|$ को संतुष्ट करने वाले बिन्दुओं z का बिन्दुपथ भी है। तब $|\alpha|^2+k^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The equation says Re $\left[(z-z_1)\overline{(z-z_2)}\right]=0$ with $z_1=1+2i$, $z_2=7+10i$, i.e. the vectors from z_1 and z_2 to z are perpendicular. So C is the circle on the segment from z_1 to z_2 as diameter: centre $c=4+6i$, radius $R=\dfrac{|z_2-z_1|}{2}=\dfrac{|6+8i|}{2}=5$.

Squaring $|z-\alpha|^2=k^2|z-\beta|^2$ (with $\beta=5+6i$) gives $(k^2-1)|z|^2-2\text{Re}\left(z\,\overline{(k^2\beta-\alpha)}\right)+k^2|\beta|^2-|\alpha|^2=0$, a circle with centre $c=\dfrac{k^2\beta-\alpha}{k^2-1}$.

Hence $\alpha-c=k^2\beta-(k^2-1)c-c=k^2(\beta-c)$ and $\beta-c=\dfrac{\beta-\alpha}{1-k^2}$. The radius is $R=\dfrac{k|\alpha-\beta|}{k^2-1}=k|\beta-c|$, and $|\alpha-c|\,|\beta-c|=R^2$ (α,β are inverse points of C).

Here $\beta-c=1$, so $k=\dfrac{R}{|\beta-c|}=5$ (and $k>1$ as required).

$\alpha=c+k^2(\beta-c)=4+6i+25=29+6i$.

Check at $z=9+6i$ (on C): $|z-\alpha|=20$, $|z-\beta|=4$, ratio 5; at $z=-1+6i$: $|z-\alpha|=30$, $|z-\beta|=6$, ratio 5.

$|\alpha|^2+k^2=(841+36)+25=902$.

Final Answer: 902

Question 21

English

Let L be the locus of the complex numbers z satisfying $|z| = |z - 2 - i|$, where $i = \sqrt{-1}$, and let R be the ray $\arg z = \dfrac{\pi}{4}$. The image of R in the line L is a ray which meets the real axis at the point $(x_0, 0)$. Then the value of x_0 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि L उन सम्मिश्र संख्याओं z का बिन्दुपथ है जो $|z| = |z - 2 - i|$ को संतुष्ट करती हैं, जहाँ $i = \sqrt{-1}$ है, तथा R किरण $\arg z = \dfrac{\pi}{4}$ है। रेखा L में R का प्रतिबिम्ब एक किरण है जो वास्तविक अक्ष को बिन्दु $(x_0, 0)$ पर मिलती है। तब x_0 का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$|z| = |z - (2 + i)|$ is the set of points equidistant from 0 and $2 + i$, the perpendicular bisector of that segment. With $z = x + iy$: $x^2 + y^2 = (x - 2)^2 + (y - 1)^2$, i.e. $4x + 2y = 5$.

The image of a point A in the perpendicular bisector of AB is B. So the image of the vertex O of the ray R (the vertex itself is not on R, but the image ray starts there) is $B = 2 + i$. Check with the image formula: the foot of O on $4x + 2y = 5$ is $\dfrac{5}{20}(4, 2) = (1, 0.5)$, and the image is twice the foot, $(2, 1)$.

R is $\{(t, t) : t > 0\}$. It meets L where $6t = 5$: $P = \left(\dfrac{5}{6}, \dfrac{5}{6}\right)$, with $t > 0$, so the ray does meet L. A point of L is its own image.

Reflection maps the ray from O through P onto the ray from B through P. Its direction is $P - B = \left(-\dfrac{7}{6}, -\dfrac{1}{6}\right)$, parallel to $(-7, -1)$.

Points of the image ray: $(2 - 7s,\ 1 - s)$ for $s > 0$. It meets the real axis when $1 - s = 0$, $s = 1 > 0$, at $x_0 = 2 - 7 = -5$.

(Direction check by reflecting the unit vector $(1, 1)/\sqrt{2}$ of R in L, normal $(2, 1)$: $(1, 1) - \dfrac{2 \cdot 3}{5}(2, 1) = \left(-\dfrac{7}{5}, -\dfrac{1}{5}\right)$, the same direction.)

Final Answer: -5

Question 22

English

Let E be the locus of the complex numbers z satisfying $10|z-1| = 3|z+\bar{z}-34|$. Then the ratio of the length of the minor axis of E to the length of its major axis is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि E उन सम्मिश्र संख्याओं z का बिन्दुपथ है जो $10|z-1| = 3|z+\bar{z}-34|$ को संतुष्ट करती हैं। तब E के लघु अक्ष की लम्बाई तथा उसके दीर्घ अक्ष की लम्बाई का अनुपात _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $z = x+iy$, $z+\bar{z} = 2x$, so $|z+\bar{z}-34| = 2|x-17|$, and the condition is $|z-1| = \dfrac{3}{10}\cdot 2|x-17| = \dfrac{3}{5}|x-17|$.

$|z-1|$ is the distance of z from the point $S = (1,0)$ and $|x-17|$ is its distance from the line $x = 17$. So E is the conic with focus S, directrix $x = 17$ and eccentricity $e = \dfrac{3}{5} < 1$: an ellipse. (Reading e as $\dfrac{3}{10}$ is the trap.)

Squaring: $(x-1)^2+y^2 = \dfrac{9}{25}(x-17)^2$, i.e. $16x^2+256x+25y^2 = 2576$, i.e. $16(x+8)^2+25y^2 = 3600$, i.e. $\dfrac{(x+8)^2}{225}+\dfrac{y^2}{144} = 1$.

So $a = 15$, $b = 12$, and $e = \sqrt{1-\dfrac{144}{225}} = \dfrac{3}{5}$, consistent with the focus-directrix form (centre -8, focus $-8+ae = 1$, directrix $-8+\dfrac{a}{e} = 17$).

Ratio of the minor axis to the major axis $= \dfrac{2b}{2a} = \dfrac{12}{15} = 0.8$.

Final Answer: 0.8

Question 23

English

Let E be the locus of the complex numbers z satisfying $|z-3-4i|+|z+3+4i| = 20$, where $i = \sqrt{-1}$. Let the latus rectum of E through the focus $3+4i$ have the end points P and Q, representing the complex numbers p and q. Then the value of $\mathrm{Re}(p) \cdot \mathrm{Re}(q)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि E उन सम्मिश्र संख्याओं z का बिन्दुपथ है जो $|z-3-4i| + |z+3+4i| = 20$ को संतुष्ट करती हैं, जहाँ $i = \sqrt{-1}$ है। माना कि नाभि $3+4i$ से होकर जाने वाले E के नाभिलम्ब के अन्त्य बिन्दु P तथा Q हैं, जो सम्मिश्र संख्याओं p तथा q को निरूपित करते हैं। तब $\mathrm{Re}(p) \cdot \mathrm{Re}(q)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The sum of distances from the fixed points $S_1 = 3+4i$ and $S_2 = -3-4i$ is $20 > |S_1 - S_2| = 10$, so E is an ellipse with foci S_1, S_2, centre 0, $2a = 20$, $2c = 10$.

So $a = 10$, $c = 5$, $b^2 = a^2 - c^2 = 75$, $e = \dfrac{1}{2}$. The major axis is along the unit vector $u = \dfrac{3+4i}{5}$.

The latus rectum through S_1 is the chord through S_1 perpendicular to the major axis, i.e. along $iu = \dfrac{-4+3i}{5}$ (multiplication by i turns through $\dfrac{\pi}{2}$). Its half-length is $\dfrac{b^2}{a} = \dfrac{75}{10} = \dfrac{15}{2}$.

Hence $p, q = S_1 \pm \dfrac{15}{2} \cdot \dfrac{-4+3i}{5} = 3+4i \pm (-6+4.5i)$, i.e. $p = -3+8.5i$ and $q = 9-0.5i$.

Check for p: $|p - S_1| = |-6+4.5i| = 7.5$ and $|p - S_2| = |0+12.5i| = 12.5$; sum $= 20$. For q: $|6-4.5i| = 7.5$ and $|12+3.5i| = \sqrt{144+12.25} = 12.5$; sum $= 20$.

(Alternatively: at an end of the latus rectum the two focal distances are $\dfrac{b^2}{a} = 7.5$ and $2a - 7.5 = 12.5$, and $7.5^2 + 10^2 = 12.5^2$, consistent with the right angle at the focus S_1.)

$\mathrm{Re}(p) \cdot \mathrm{Re}(q) = (-3)(9) = -27$.

Final Answer: -27

Question 24

English

Let z and w be complex numbers satisfying $z + (1+i)\bar{w} = 1 + i$ and $2\bar{z} + (1-i)w = 1$, where $i = \sqrt{-1}$. Then the value of $\text{Im}(zw)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि z तथा w ऐसी सम्मिश्र संख्याएँ हैं जो $z + (1+i)\bar{w} = 1 + i$ तथा $2\bar{z} + (1-i)w = 1$ को संतुष्ट करती हैं, जहाँ $i = \sqrt{-1}$ है। तब $\text{Im}(zw)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The unknowns appear as $z, \bar{w}$ in the first equation and as $\bar{z}, w$ in the second. Taking the conjugate of the second equation (the conjugate of a sum and product is the sum and product of conjugates): $2z + (1+i)\bar{w} = 1$.

Now both equations are linear in z and $\bar{w}$: $z + (1+i)\bar{w} = 1 + i$ and $2z + (1+i)\bar{w} = 1$.

Subtracting the first from the second: $z = 1 - (1+i) = -i$.

Then $(1+i)\bar{w} = 1 + i - (-i) = 1 + 2i$, so $\bar{w} = \dfrac{1+2i}{1+i} = \dfrac{(1+2i)(1-i)}{2} = \dfrac{3+i}{2}$ and $w = \dfrac{3-i}{2}$.

Check in the original second equation: $2\bar{z} + (1-i)w = 2i + \dfrac{(1-i)(3-i)}{2} = 2i + \dfrac{2-4i}{2} = 2i + 1 - 2i = 1$.

$zw = -i \cdot \dfrac{3-i}{2} = \dfrac{-3i + i^2}{2} = -\dfrac{1}{2} - \dfrac{3}{2}i$, so $\text{Im}(zw) = -1.5$.

Final Answer: -1.5

Question 25

English

Let S be the set of all complex numbers z satisfying $z^5 = 81\bar{z}$. Then the value of $\sum_{z \in S} |z - 3|^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी सम्मिश्र संख्याओं z का समुच्चय है जो $z^5 = 81\bar{z}$ को संतुष्ट करती हैं। तब $\sum_{z \in S} |z - 3|^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

$z = 0$ satisfies $z^5 = 81\bar{z}$, so $0 \in S$.

For $z \neq 0$, taking moduli: $|z|^5 = 81|z|$, so $|z|^4 = 81$, $|z| = 3$.

Then $z\bar{z} = 9$, so $\bar{z} = \dfrac{9}{z}$ and the equation becomes $z^5 = \dfrac{729}{z}$, i.e. $z^6 = 729 = 3^6$. Conversely each sixth root of 729 has modulus 3 and satisfies the equation.

So $S = \{0\} \cup \left\{3e^{ik\pi/3} : k = 0, 1, \ldots, 5\right\}$, seven elements.

For $|z| = 3$: $|z - 3|^2 = |z|^2 - 6\,\mathrm{Re}\, z + 9 = 18 - 6\,\mathrm{Re}\, z$. The six roots sum to 0, so $\sum \mathrm{Re}\, z = 0$ and these contribute $6 \cdot 18 = 108$.

(Directly: the distances from 3 are $0, 3, 3\sqrt{3}, 6, 3\sqrt{3}, 3$, with squares $0 + 9 + 27 + 36 + 27 + 9 = 108$.)

The element $z = 0$ adds $|0 - 3|^2 = 9$. Total $= 117$.

Final Answer: 117

Question 26

English

Let C be the circle $z\bar{z} + (3-4i)z + (3+4i)\bar{z} + 21 = 0$ in the Argand plane, where $i = \sqrt{-1}$, and let K be its centre. The line $(2-i)z + (2+i)\bar{z} + 12 = 0$ meets C at the points P and Q. Then the value of $\cos\angle PKQ$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि C आर्गण्ड तल में वृत्त $z\bar{z} + (3-4i)z + (3+4i)\bar{z} + 21 = 0$ है, जहाँ $i = \sqrt{-1}$ है, तथा K इसका केन्द्र है। रेखा $(2-i)z + (2+i)\bar{z} + 12 = 0$, C को बिन्दुओं P तथा Q पर मिलती है। तब $\cos\angle PKQ$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A circle $z\bar{z} + \bar{a}z + a\bar{z} + b = 0$ (b real) can be written $|z+a|^2 = |a|^2 - b$. Here the coefficient of $\bar{z}$ is $a = 3+4i$ and that of z is $\bar{a} = 3-4i$, so $|z+3+4i|^2 = 25 - 21 = 4$.

So $K = -3-4i$ and $r = 2$.

For $z = x+iy$, $(2-i)z + (2+i)\bar{z} = 2\,\mathrm{Re}((2-i)(x+iy)) = 2(2x+y)$, so the line is $4x+2y+12=0$, i.e. $2x+y+6=0$.

Distance of $K(-3,-4)$ from the line: $d = \dfrac{|-6-4+6|}{\sqrt{5}} = \dfrac{4}{\sqrt{5}} < 2$, so the line cuts C at two points P, Q.

In the isosceles triangle PKQ the perpendicular from K bisects $\angle PKQ$, so $\cos\dfrac{\angle PKQ}{2} = \dfrac{d}{r} = \dfrac{2}{\sqrt{5}}$.

$\cos\angle PKQ = 2\cdot\dfrac{4}{5} - 1 = \dfrac{3}{5} = 0.6$.

Check: the line meets C at $P = -1-4i$ and $Q = -\dfrac{9}{5} - \dfrac{12}{5}i$; $\overrightarrow{KP} = (2,0)$, $\overrightarrow{KQ} = \left(\dfrac{6}{5}, \dfrac{8}{5}\right)$, and $\cos\angle PKQ = \dfrac{12/5}{2\cdot 2} = 0.6$.

Final Answer: 0.6